Observe, Imagine, Abstain: Detecting Hallucinated Geometry in Feed-Forward 3D Reconstruction

Anonymous CVPR submission

Abstract

*Feed-forward 3D foundation models reconstruct a scene from unposed photographs in a single pass. Where no photograph shows a surface, they fill the gap from learned priors, and they often assign the invented geometry a confidence indistinguishable from that of observed geometry. We call this geometric hallucination and define it operationally: a predicted point is hallucinated when it lies on a surface that no input view observes. HalluBench-3D produces such labels for free by hiding a 3D region of the ground-truth surface in every input view, so that every pixel inside the region is unsupported and every pixel far from it is supported. We ask whether a model's native confidence separates the two (*AUROC*) and how much unsupported geometry survives a threshold that keeps 90% of supported geometry, the Confident-Unsupported Rate (*CUR*). Across five models (VGGT, Pi3, Pi3X, MapAnything, Depth Anything 3), 18 DTU scans and 18 CO3Dv2 scenes, native confidence lets between 24% and 81% of unsupported pixels through. A realistic inpainted fill makes leakage worse, dropping views or adding a distractor leaves it unchanged, variance over perturbed inputs is worse than native confidence, and the ranking of models changes between the lab rig and phone captures. We then ask what a detector of unobserved geometry can be. A small* support head *on the frozen multi-view features of the backbone predicts, per pixel, whether the output has multi-view support, and abstains in between. It needs no ground truth: its labels are the support count of the backbone's own prediction, the quantity a reprojection test measures, and the head learns to read it from the features in one forward. On four backbones (VGGT, Pi3X, Pi3, Depth Anything 3) and four datasets its training never touched (DTU benchmark scans with 3 to 10 clean views, CO3Dv2 phone orbits, ETH3D building scans, Internet photographs of nine heritage landmarks) it separates single-view from supported geometry at* AUROC *0.93 to 0.99 against 0.55 to 0.89 for native confidence, lets 2 to 32 times fewer unsupported pixels through a threshold that keeps 90% of supported ones, matches a head trained on ground-truth visibility within 0.02, and beats a per-image control that falls below chance off the lab rig. It beats reprojection consistency on DTU, CO3Dv2 and the landmark photographs and ties it on ETH3D, at a fraction of the cost. Two cautions: a head trained on the benchmark's fills learns the fill, and no detector, this head included, fires on a filled patch that every view shows; five models of three families invent the same surface there, so disagreement between models does not find it either; what does move it is the fill itself, so the invented surface is read from the hole's pixels rather than fixed by the prior. Code, benchmark, heads and the full run log will be released.*

1. Introduction

Feed-forward 3D reconstruction has moved from pairwise pointmaps [14, 31] to models that take dozens of unposed images and return dense geometry in one pass [13, 17, 29, 32]. They are trained to output geometry at every pixel, including pixels whose true surface no input view shows; the objective rewards a plausible guess, so they learn to guess. The trouble is that the confidence map, the one signal a user has, does not say which parts were observed and which were guessed.

Suppose you are a conservator looking at a reconstruction of a dome, and you see a crack. Before you send someone up a scaffold, you want to know whether any photograph shows it or whether the model drew it from what domes usually look like. Today you cannot tell: the confidence map is bright in both cases.

We make three contributions.

1. **A definition and a benchmark.** We define hallucinated geometry as predicted points on surfaces that no input view shows. HalluBench-3D produces these labels by 3D-consistent masking of ground-truth surfaces, with controls on the same pixels under clean input. We report AUROC, CUR and CUR-ERR for five models on DTU and CO3Dv2, three fill types, an inpainted fill, and stress tests (view dropping, low parallax, a distractor image), each with cluster-bootstrap intervals and paired permutation tests.
2. **The backbone already knows, and a label-free way to read it.** Multi-view support is linearly decodable from the second half of the backbone's cross-view layers and

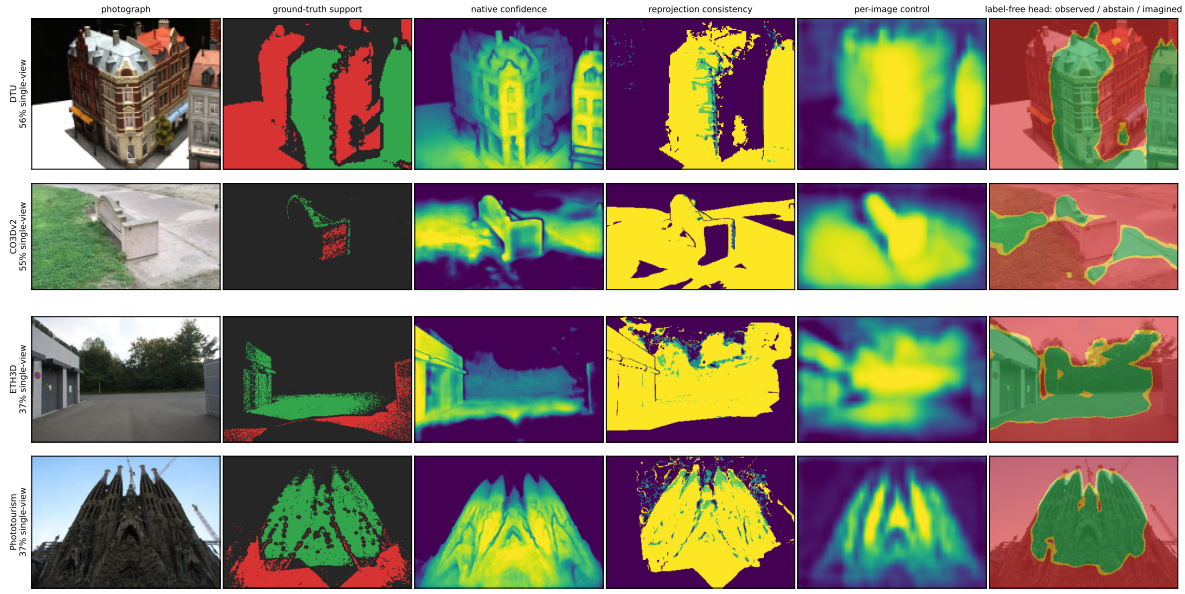


Figure 1. **What the label-free head says, on four captures it never saw.** One view of a clean multi-view input per dataset (DTU rig, CO3Dv2 phone orbit, ETH3D DSLR, a tourist photograph of the Sagrada Família); ground-truth support (red: no other input view sees the surface; green: two or more do; dark: no ground truth); VGGT's native confidence; reprojection consistency with its own cameras; the per-image control; and the head's bands at the fixed thresholds (green observed, yellow abstain, red imagined). The head's labels came from the backbone's own predictions on DTU training scans. Where native confidence lets half of the single-view pixels through a threshold that keeps 90% of the supported ones on DTU, the head lets 1.7% through. Rendered from the released per-pixel scores (run P6-dump).

absent from its per-image tokens; a linear probe gets within 0.015 of any head we trained. We read that signal with a small head whose labels come from the backbone's own prediction, so it trains on any capture with no ground truth, and abstains in between. On four backbones and four datasets it never saw it passes two pre-registered tests against native confidence, a per-image control and a ground-truth-trained head (Section 7). Trained on the benchmark's synthetic fills instead, it learns the fill, which we report as a warning for masking-based supervision.

3. **Controls that a detector must pass, and one it fails.** Native confidence, perturbation variance, reprojection consistency, two hand-crafted cues, a linear probe and a head that sees one image at a time are scored on identical rows with paired tests. Put back to the benchmark's own filled patches, the head does not fire: it certifies agreement between views, which a consistent invention has.

Every number comes from a logged run with a run id; the pass conditions of Section 7 were written down, with dates, before the heads they judge were trained (supplement). The headline claim we test, and do not assume, is that native confidence fails to flag unsupported geometry across current models; it holds, and more strongly on phone captures than on the lab rig.

2. Related work

Feed-forward 3D reconstruction. DUSt3R and MASt3R [14, 31] predict pointmaps for image pairs. VGGT [29], Pi3 and Pi3X [32], MapAnything [13], VGGT-Omega [30] and Depth Anything 3 [17] extend this to many unposed views in one pass. Each exposes a per-pixel confidence trained as a loss weight on observed data; none is trained to distinguish observed from unobserved surfaces. Pi3X replaces Pi3's near-binary confidence with continuous quality levels; MapAnything also offers a multi-view consistency confidence, which we include as a baseline.

Hallucination in feed-forward 3D. Bratulic et al. [3] probe DUSt3R, VGGT and DA3 and show that attention heads keep matching correspondences after small patches are occluded, concluding that these models hallucinate missing correspondences from learned priors; they measure attention and Sampson error on ShapeNet pairs, not confidence or 3D labels. Paul et al. [23] benchmark hallucination at the input level: unrelated scenes, repeated images and noise still produce dense geometry, scored by COLMAP-based consistency. 3D-Mirage [21] studies monocular depth under illusions; Li et al. [15] show that wide viewpoint spans produce unsupported geometry, and Dadajonov and Umarhojiyev [5] that unobserved parts of a room are not revealed by inference-time uncertainty; neither has per-point labels.

FreeInpaint [22] fills masked regions on purpose, the mechanism these models apply unasked. Our definition is per point, with exact supported and unsupported labels, and it tests whether native confidence separates them.

Uncertainty for feed-forward 3D. ReVar3R [18] takes per-point variance over perturbed runs; it beats native confidence on AUSE in most conditions but, by its authors' account, misses stable systematic bias, which is what a confidently filled region is. Trust3R [33] fits an evidential head on a frozen backbone, GeoCond [1] adds pose-level uncertainty, and Deng et al. [7] find the implied uncertainty of seven backbones at least 2.4× too small. All score uncertainty against ground-truth error, which conflates confident errors on observed surfaces with confident predictions on unseen surfaces; CUR isolates the second case. Closest to our detector, a linear probe on frozen VGGT features predicts *camera-pose* failure with AUROC 0.935 against 0.797 for native confidence [28], and a small head on the same features predicts pairwise co-visibility [34]; our support head is the dense, per-pixel analogue for unobserved *geometry*, trained either on ground-truth visibility or on the backbone's own multi-view agreement with no label at all, and the linear probe is one of our controls.

Generative and multi-sample 3D. Z3D [2] decodes hidden surfaces from frozen VGGT features, evidence that unobserved geometry is present in them. Feed-forward Gaussian splatting from unposed views [9] outputs pixel-aligned Gaussians with opacities, and ReconSplat [26] extrapolates beyond the observed views by design; a model built to extrapolate has all the more reason to say where it did. World models that keep a persistent Gaussian memory of the frames they generate [4] store imagined and observed surfaces alike, which is where a per-point support signal belongs. VolFill [20] finds sample variation 1.3× larger in occluded regions; diffusion depth ensembles [11, 12] give sample-disagreement uncertainty; a generative head over quantized geometry [6, 19] whose samples disagree where the model imagines is the natural successor to our discriminative head, and future work.

Hallucination in vision-language models. POPE [16] labels statements unsupported by the image; ours is the geometric analogue, geometry unsupported by any view.

3. Defining geometric hallucination

Given input views $I_1, \ldots, I_N$, a model returns for each view i a depth map D_i and a confidence map C_i. A pixel p in view i is *supported* if its ground-truth surface point is visible in at least one input view, and *unsupported* if it is visible in none. A model hallucinates at p when it outputs geometry at an unsupported p with high confidence.

Why confidence is the wrong tool. Native confidence is trained as a per-pixel loss weight on observed data: low where the regression target is noisy (edges, specular surfaces), high where it is smooth. A region that no view sees and that the prior fills smoothly therefore receives high confidence by construction; Section 5 measures how large the effect is.

Metrics. All are computed per (model, scene, seed, radius, fill) and averaged with cluster-bootstrap intervals over scenes. Pixels whose confidence ties with τ count as kept, so a near-binary confidence (Pi3) is not favoured by the threshold.

- AUROC: $P(C_S > C_U)$ with ties counted half. 1.0 means every supported pixel outranks every unsupported one; 0.5 means confidence carries no information about support.
- CUR: choose τ so that 90% of supported pixels have $C \geq \tau$; CUR is the fraction of unsupported pixels with $C \geq \tau$. This is the hallucination rate you would experience after filtering by confidence.
- CUR-ERR: the fraction of unsupported pixels above τ whose AbsRel exceeds 0.1. Confident and wrong.
- AbsRel_U, AbsRel_S: depth error on unsupported and supported pixels after median scaling on supported pixels; controls: the same pixels under clean input (AbsRel_U ctrl) and the ratio of median confidence inside the hidden region to its clean value.

4. HalluBench-3D

Protocol. For each scene we take 10 views. A sphere is centred on a random ground-truth surface point in the reference view, with radius 0.10 or 0.18 of the object's bounding-box diagonal, resampled until it covers 2 to 35% of the reference view. Every pixel in every view whose ground-truth point lies inside the sphere is hidden (the protocol figure in the supplement). Hidden pixels are filled with gray, a blur of the original, uniform noise, or an inpainting of the surround [27]; fill type is an ablation so that results are not an artefact of a visible patch. Labels: U is hidden in all views; S is valid and outside a 6-pixel dilation of the hidden region; the ring in between is ignored. Five mask seeds per scene; masks are deterministic in (scene, seed, radius).

Datasets. *DTU* [8]: the 18 scans of the MVSNet validation split, 10 views spread over the 49-camera rig, model input 420 × 518. *CO3Dv2* [24]: 18 scenes, the two depth-carrying sequences of each of nine categories (apple, ball, bench, hydrant, mouse, teddybear, toytrain, toytruck, vase) from the single-sequence subset, with the official COLMAP depth as ground truth. The 10 views are spread over the

full orbit, so the hidden region is seen from the back as well as the front. Frames are 640×360 phone video; models run at 518×294 (or the transpose for landscape sequences). Everything else, including every threshold, is unchanged from DTU. *ETH3D* [25]: nine DSLR building-scale scenes with laser-scanner ground truth on the undistorted pinhole frames (640×427, models 518×350), the reference frame and its nine most co-visible frames. *Phototourism* [10]: nine landmarks of the Image Matching Challenge 2020 set (British Museum, Florence cathedral, Lincoln Memorial, London Bridge, Milan cathedral, Mount Rushmore, Piazza San Marco, Sagrada Familia, St Paul's): Internet photographs with people and vehicles in front of the monument, unordered viewpoints and focal lengths, COLMAP multi-view-stereo depth as ground truth; per landmark the best-connected photograph and its 19 most co-visible neighbours at 640×448; all nine pass the correctness guards (supplement). It is used only by the true-visibility protocol of Section 7, which draws its own view sets (3, 5 and 10 clean views, three rotations of the set) on every dataset. Training-data overlap: VGGT, Pi3/Pi3X and Depth Anything 3 list CO3Dv2 among their training sets [17, 29, 32]; MapAnything does not [13]; none of the four lists DTU or ETH3D. CO3Dv2 is therefore in-distribution for three of the backbones, and DTU and ETH3D are out of distribution for all of them.

Models. VGGT-1B [29] (depth and depth confidence), Pi3 and Pi3X [32] (local point z and sigmoid confidence), MapAnything [13] (image-only metric mode), and Depth Anything 3 [17] (Large 1.1). Same views and resolution for all; fp16 on a T4. A feed-forward Gaussian-splatting model, AnySplat [9], scored on each pixel's Gaussian depth with its opacity as confidence, is in the supplement from a partial run: the representation does not change the answer.

Statistics. Means carry 95% cluster-bootstrap intervals over scenes (10,000 resamples; each scene contributes its mean over seeds). Pairwise comparisons use a paired sign-flip permutation test on per-scene means (20,000 permutations), Holm-corrected within each family of pairs; the smallest p the test can return is 0.00005, written $p < 0.0001$ below.

5. Results on HalluBench-3D

5.1. DTU: native confidence leaks

Table 1 pools the full DTU configuration over the two natural-statistics fills, blur and noise (per-fill tables and the per-run spread in the supplement): CUR is 0.242 for VGGT, 0.281 for Pi3, 0.532 for Pi3X, 0.622 for MapAnything and 0.701 for Depth Anything 3, with AUROC 0.903, 0.912, 0.813, 0.743 and 0.711. Three tiers emerge. VGGT and Pi3 drop their confidence inside the hidden region (to 19 to 41% of its clean value) yet still let one in four unsupported pixels through a threshold that keeps 90% of supported ones. Pi3X leaks one in two, and under a noise fill it is as confident about geometry it cannot see as about geometry it can (ratio 1.09). MapAnything and Depth Anything 3 are close to blind to non-observation: DA3's confidence inside the hidden region stays at 80 to 86% of its clean value for every fill, including the gray patch.

Pairwise, after Holm correction over the ten pairs: VGGT and Pi3 are indistinguishable ($p = 0.41$); both leak less than Pi3X ($p \leq 0.0015$), MapAnything and DA3 ($p = 0.0005$). Pi3X versus DA3 ($p = 0.052$), Pi3X versus MapAnything and MapAnything versus DA3 ($p = 0.19$) are not separated.

Confident and wrong. On DTU the imagined depth is accurate for four of the five models (ABSREL$_U$ 0.011 to 0.022, about 1.3 to 2× the clean value), so CUR-ERR stays near zero. The failure is that you cannot tell imagined from observed, not that the imagination is poor. MapAnything is the exception: 29% of its unsupported pixels are confident and wrong by more than 10% ABSREL, on top of a 10% error on supported pixels in image-only metric mode (unchanged at its native input size). The pass condition fixed before the benchmark was scaled (CUR above zero with intervals excluding zero, AUROC clearly below one on the natural fills) holds for all five models.

5.2. Stress tests: the leak is not a patch artefact

A reasonable objection to Table 1 is that a filled patch is visible, and the model is reacting to the patch rather than to the absence of a surface. The stress table in the supplement answers it four ways. An inpainted fill, edge-continuous and without texture discontinuity, raises leakage further for VGGT ($+0.088$, $p = 0.031$) and Pi3X ($+0.068$, $p = 0.005$), so Table 1 is a lower bound on what a natural-looking occluder produces. Dropping from 10 to 3 views leaves CUR unchanged for VGGT and Pi3X ($p > 0.4$; DA3 moves at 5 views, $+0.11$, $p = 0.008$, and not at 3, which we read as noise): fewer views to cross-check does not make the models less sure, so native confidence is driven by single-view appearance rather than multi-view agreement. With 3 views or low parallax, Pi3X's imagined depth degrades (ABSREL$_U$ 0.05, CUR-ERR up to 0.04) while its confidence does not move: the confident-and-wrong fraction, invisible on the easy rig, appears under harder capture. DA3 is different: a distractor image raises its leakage and low-parallax input halves it, consistent with a confidence that tracks appearance consistency across views rather than observation of the surface, an observation we flag rather than explain.

Table 1. **Native confidence leaks, on the lab rig and on phone captures.** Five models, full configuration (18 scenes × 5 mask seeds × 2 radii), pooled over the blur and noise fills; mean [95% cluster-bootstrap CI over scenes]. *conf ratio*: median confidence inside the hidden region over its clean value. Per-fill tables with every column in the supplement. Runs P1-full, P2-full, P2-co3d, P2-co3d-b; T4, fp16.

model	DTU, 18 scans				CO3Dv2, 18 scenes			
	AUROC	CUR	CUR-ERR	conf ratio	AUROC	CUR	CUR-ERR	conf ratio
VGGT-1B	0.903 [0.873, 0.929]	0.242 [0.181, 0.304]	0.000	0.22	0.681 [0.632, 0.729]	0.651 [0.566, 0.734]	0.031	0.70
Pi3	0.912 [0.881, 0.940]	0.281 [0.189, 0.379]	0.002	0.34	0.891 [0.845, 0.930]	0.272 [0.190, 0.360]	0.002	0.43
Pi3X	0.813 [0.762, 0.860]	0.532 [0.403, 0.663]	0.003	0.91	0.732 [0.680, 0.779]	0.570 [0.500, 0.641]	0.001	0.87
MapAnything	0.743 [0.687, 0.795]	0.622 [0.534, 0.707]	0.286	0.69	0.576 [0.521, 0.632]	0.810 [0.756, 0.863]	0.007	0.68
Depth Anything 3	0.711 [0.653, 0.763]	0.701 [0.614, 0.784]	0.002	0.83	0.620 [0.567, 0.670]	0.797 [0.728, 0.861]	0.000	0.88

5.3. CO3Dv2: the ranking changes

Table 1 also gives CO3Dv2, the full protocol on phone captures with every threshold frozen from DTU. Pooled over blur and noise, CUR is 0.272 for Pi3, 0.570 for Pi3X, 0.651 for VGGT, 0.797 for DA3 and 0.810 for MapAnything; AUROC is 0.891, 0.732, 0.681, 0.620 and 0.576.

The ranking changes. Pi3 is the only model whose behaviour transfers from the lab rig to phone video (CUR 0.27 on both). The leak is larger on the dataset three of the models were trained on: seeing CO3Dv2 at training time did not teach their confidence to notice a missing surface. VGGT, joint best on DTU, leaks 65% of unsupported pixels on CO3Dv2, 2.7× its DTU value, and under a noise fill its confidence is almost blind to non-observation (AUROC 0.59). MapAnything and DA3 let about four in five unsupported pixels through. After Holm correction, Pi3 leaks less than every other model ($p = 0.0005$ each); Pi3X leaks less than MapAnything and DA3 ($p = 0.0005$); VGGT leaks less than MapAnything ($p = 0.010$) and DA3 ($p = 0.0005$); VGGT versus Pi3X ($p = 0.23$) and MapAnything versus DA3 ($p = 0.73$) are not separated. Across datasets, the intervals of VGGT, MapAnything and DA3 do not overlap; those of Pi3 and Pi3X do.

Imagination is accurate here too. ABSREL_U is within 0.005 of ABSREL_S for every model, so CUR-ERR stays below 0.04, and MapAnything's confident-and-wrong behaviour from DTU does not appear (ABSREL_S 0.041, CUR-ERR 0.007). These are everyday objects with strong shape priors; the models complete them well and report the completion with the confidence of an observation.

Fills and scenes. On DTU every fill raised CUR over gray for VGGT, Pi3 and Pi3X. On CO3Dv2 noise raises VGGT's CUR by 0.20 ($p < 0.0001$) but blur does not ($p = 0.30$); Pi3X reacts to both (+0.26, +0.23), Pi3 weakly, MapAnything and DA3 not at all (p from 0.06 to 0.91): a confidence that does not change when the patch changes is not reading the patch. The leak is not driven by a few scenes (per-scene heatmap in the supplement), and removing the one scene that missed the mask-transfer guard moves every pooled number by at most 0.013.

5.4. Cheap detectors do not help

Two obvious alternatives, scored on the DTU rows with blur, radius 0.18 and seed 0 (table in the supplement). Variance over $K = 4$ photometrically perturbed runs is much worse than native confidence on both backbones (AUROC 0.67 and 0.55 against 0.92 and 0.83, paired $p \leq 0.0007$) at four times the cost: the imagined geometry is stable under input perturbation, the systematic bias ReVar3R's authors describe. Reprojection consistency with the model's own cameras is no better than chance (AUROC 0.49, 0.48): the imagined surface is cross-view coherent. MapAnything's multi-view consistency confidence leaks more than its learned confidence (AUROC 0.67, CUR 0.78 against 0.74 and 0.62 on the same rows, run P6-mapanything-mv).

6. Method: observe, imagine, abstain

The benchmark says that native confidence does not know what the model has seen. Rather than suppress the prior, which is what makes these models useful, we want the model to say which of its outputs came from the prior.

Architecture. A frozen backbone (VGGT, Pi3X, Pi3 or Depth Anything 3) provides geometry, native confidence, and the multi-view features that produced them: for VGGT and DA3 the tokens of the four cross-view layers their depth heads read (2048-d at 14×14 patches), for Pi3X and Pi3 the 2048-d decoder tokens feeding their confidence heads. The *support head* (LayerNorm and 1×1 projection per layer to 256 channels, sum, two 3×3 convolutions at patch resolution, one logit per patch upsampled bilinearly) outputs $P(\text{imagined})$; the benchmark scores $1 - P(\text{imagined})$. 3.3 M parameters on VGGT and DA3, 1.7 M on Pi3X and Pi3; the backbone is neither modified nor re-run.

Training labels. Three label sources, all on the 79 scans of the DTU training split, disjoint from the 18 benchmark scans and the 22 extension scans of Section 7; the views are clean, 3, 5 or 10 at random per step. *Self-support labels* (the main head, label-free): the backbone is run once; each pixel's *predicted* point is projected with the *predicted* depth and cameras into the other input views and counted as supported

where it lands on their predicted surface (3% depth tolerance, the 20% least confident pixels dropped). A pixel no other view supports is the positive class, a pixel supported by two or more views outside a 3-pixel margin the negative class. This is the reprojection test of Section 5.4 used as a teacher, so the head has to generalise beyond it to be worth having. *Visibility labels* (the upper bound): the same count from ground-truth depth and cameras. *Fill labels* (the cautionary variant): a sphere placed on the predicted surface is hidden in every view (gray, blur or noise; inpaint held out) and the head learns to find the hidden pixels, with 15% clean samples. All three: class-balanced binary cross-entropy, 1500 AdamW steps (learning rate 10^{-3}, cosine, warm-up 100), one scene of random views per step, 10 minutes per backbone on one H200; hyperparameters fixed before any benchmark scan was evaluated.

Inference and compute. One backbone forward, one head forward; the head adds under 1% of the backbone's FLOPs. Three colours for the user: *observed* above $1 - P(\text{imagined}) = 0.7$, *imagined* below 0.3, *abstain* between, fixed in advance; the metrics use the continuous score.

Controls and ablations. A head could learn an image cue (the patch; image borders and depth edges, where single-view pixels concentrate) rather than the missing support. The decisive control is the *per-image feature head*: the same head on per-image encoder tokens before any cross-view attention, same labels. Two hand-crafted scores test the same objection without training: the distance to the image border and minus the gradient of the predicted depth. Further controls: a *linear probe* on the same features [28] and on each cross-view layer alone, two extra initialisation seeds, the held-out inpainted fill and CO3Dv2 for the fill-trained head; heads sharing a label source train jointly on the same backbone forwards.

7. Experiments on the method

Fill-trained head: what it learns. Its pre-registered test asked it to beat native confidence ($p < 0.05$) and its per-image control on the held-out inpainted fill and on CO3Dv2. It scores AUROC 0.990 to 1.000 on every cell of the full configuration and the stress tests, and so does the control (1.000 against 0.999 for VGGT and 0.994 against 0.994 for Pi3X on the inpainted fill; 0.992 against 0.989 and 0.969 against 0.988 on CO3Dv2); its formal clause passed on a 0.003 difference, a flaw of the criterion. Put to real single-view geometry (supplement), the same head scores *below* native confidence on every dataset: it learned the fill and nothing else, and the per-image control is mandatory for anyone training on masking labels.

What the tokens know. The per-image control falls below chance on CO3Dv2 and ETH3D, and so does the distance to the image border (indistinguishable from the control there, $p > 0.3$): a head that cannot see across views learns the frame edge, which marks single-view pixels on the rig and not elsewhere. A linear probe on the same cross-view features scores within 0.015 of the convolutional head of the next paragraphs on every dataset, so the information is in the features and any head is a reader. A probe on one VGGT layer at a time says where it appears (DTU / CO3Dv2): layer 4 of 24 scores 0.87 / 0.48, like the per-image control, layer 11 0.95 / 0.70, layer 17 0.99 / 0.92, layer 23 0.99 / 0.88: support becomes linearly decodable in the second half of the aggregator.

Visibility head: the pre-registered test. Gate 3 was written before any visibility head was trained, with a margin this time: beat native confidence on AUROC and CUR ($p < 0.05$) and the per-image control on AUROC by at least 0.02 ($p < 0.05$), on DTU and on one unseen dataset. Table 2: the ground-truth-trained head reaches AUROC 0.990 to 0.993 on DTU against 0.77 to 0.84 for native confidence and 0.86 to 0.88 for the control (supplement table), $+0.11$ to $+0.13$ at every view count; off the rig it keeps 0.93 to 0.98 while the control falls *below chance* on CO3Dv2. Every paired difference is at the permutation floor ($p < 0.0001$ on 18 scenes, 0.0043 on nine); Gate 3 passes on three datasets and three backbones.

Label-free head: the method. Gate 4, written before any label-free head was trained, asks the same of the head trained on the backbone's own support count, and adds: within 0.03 AUROC of the ground-truth-trained head on DTU, and ahead of reprojection consistency, its own teacher, on CO3Dv2 at $p < 0.05$. It passes on all three backbones of Table 2, and on Depth Anything 3, the worst leaker of Table 1, where the label-free head scores 0.98 / 0.88 / 0.97 / 0.93 on DTU / CO3Dv2 / ETH3D / landmarks against 0.84 / 0.68 / 0.65 / 0.78 for native confidence and 0.76 / 0.44 / 0.75 / 0.73 for its per-image control (supplement). The label-free head is within 0.02 of the ground-truth-trained head on every dataset, beats its per-image control by 0.10 to 0.12 on DTU and 0.35 to 0.50 elsewhere, and beats reprojection consistency by 0.04 to 0.07 on DTU and by 0.14, 0.05 and 0.04 on CO3Dv2 ($p < 0.0001$, 0.006, 0.008); on ETH3D the two tie (± 0.02, $p > 0.15$): with adjacent DSLR views of a building the reprojection test is as good as any head, and the head's advantage there is cost alone.

Tourist photographs, adaptation, more views. The landmark photographs are the capture the detector is meant for and the hardest for the alternatives: people stand between camera and monument and focal lengths differ from photo

Table 2. **True-visibility protocol: detecting single-view geometry from clean views.** Pooled AUROC and CUR over scenes and view counts (3, 5 and 10 views; 3 and 5 on CO3Dv2, where ten views around an object leave no single-view pixel). No head saw any of these scenes; the last row per backbone is the label-free head trained on the other half of the target dataset instead of DTU. Runs P4-vis*, P7-vis-imc, P6-adapt-*, P7-adapt-imc. DA3 and MapAnything native confidence on the same rows: AUROC 0.836 / 0.675 / 0.645 / 0.782 and 0.848 / 0.719 / 0.789 / 0.893. Intervals, per-view-count tables and paired tests: supplement.

detector	DTU		CO3Dv2		ETH3D		Photot.	
	AUROC	CUR	AUROC	CUR	AUROC	CUR	AUROC	CUR
VGGT-1B								
native confidence	0.772	0.530	0.754	0.723	0.715	0.566	0.802	0.451
reprojection consistency	0.954	0.082	0.793	0.693	0.975	0.069	0.824	0.340
support head, GT visibility labels	0.993	0.015	0.927	0.211	0.980	0.055	0.908	0.258
per-image control (self labels)	0.885	0.325	0.460	0.892	0.626	0.750	0.729	0.681
support head, label-free	0.992	0.017	0.931	0.199	0.980	0.051	0.930	0.191
label-free head, adapted on the target domain	–	–	0.841	0.474	–	–	0.926	0.203
Pi3X								
native confidence	0.820	0.493	0.697	0.846	0.887	0.304	0.839	0.450
reprojection consistency	0.914	0.164	0.885	0.224	0.977	0.054	0.814	0.361
support head, GT visibility labels	0.990	0.021	0.930	0.221	0.975	0.072	0.938	0.176
per-image control (self labels)	0.871	0.363	0.469	0.877	0.610	0.776	0.719	0.697
support head, label-free	0.988	0.025	0.939	0.190	0.963	0.103	0.940	0.166
label-free head, adapted on the target domain	–	–	0.964	0.094	–	–	0.929	0.198
Pi3								
native confidence	0.841	0.489	0.551	0.882	0.814	0.498	0.847	0.447
reprojection consistency	0.952	0.089	0.894	0.205	0.976	0.039	0.817	0.345
support head, GT visibility labels	0.992	0.016	0.934	0.203	0.953	0.135	0.932	0.201
per-image control (self labels)	0.888	0.320	0.471	0.895	0.654	0.750	0.724	0.683
support head, label-free	0.989	0.021	0.930	0.225	0.951	0.139	0.931	0.181

to photo, so the model's own cameras are least reliable and reprojection consistency scores 0.81 to 0.82. The label-free head scores 0.93 to 0.94 on all three backbones against 0.80 to 0.85 for native confidence (paired $p \leq 0.016$) and 0.72 to 0.73 for the per-image control; it beats reprojection consistency by 0.11 to 0.13 ($p = 0.0043$, the nine-scene floor). Trained on the target capture itself, still without labels (one half of a dataset, scored on the other), the head gains where the backbone's own geometry on that capture is reliable, Pi3X on CO3Dv2 ($+0.025$, $p = 0.007$) and on ETH3D ($+0.027$, $p = 0.008$), changes nothing on the landmarks (± 0.01, $p > 0.4$), and hurts VGGT on CO3Dv2 (-0.09, $p = 0.0004$), whose reprojection test, the teacher, scores 0.79 there against 0.89 for Pi3X. The DTU-trained head is the safe default and never touches the target capture.

What abstaining buys. CUR sets its threshold on the test scene itself, which a deployed filter cannot do. From the per-pixel scores (run P6-dump; supplement): the area under the single-view rate of the kept pixels against coverage is within 0.002 to 0.02 of the oracle for the label-free head on every dataset and two to four times larger for native confidence; keeping the half the head trusts most lowers the depth error of what remains by 15 to 44% (native confidence lowers it as much: it tracks error, the head tracks support). With the bands 0.3 and 0.7 fixed in advance, the head marks 57 to 87% of the labelled pixels *observed*, of which 0.5 to 5% are single-view (per-image control: 50 to 94% with 4 to 20%). Downstream, fusing the views with the backbone's own cameras and keeping the 80% of points the head trusts most raises the accuracy of the fused cloud on DTU from 0.65 to 0.77 (native confidence 0.78, reprojection 0.76, per-image control 0.70), while the oracle that drops single-view pixels first reaches 0.66: on DTU single-view geometry is nearly as accurate as supported geometry, so abstaining buys corroboration, not accuracy (ETH3D is the same picture; supplement). Multiplying the head by native confidence, one score for error and one for support, changes the head's AUROC by -0.055 to 0.004 over the four backbones and four datasets, and in the fused cloud at 0.8 coverage reaches accuracy 0.79 (head 0.77, native 0.78) at a single-view rate of 4% (native 5%): the product inherits the accuracy filter of native confidence and keeps most of the head's corroboration, and neither signal substitutes for the other. A threshold set on the *other* scenes of a dataset lets 1 to 2% of single-view pixels through on DTU, 7 to 16% on CO3Dv2 and ETH3D and 17 to 20% on the landmarks, against 57 to 83% for native confidence, and a DTU threshold applied unchanged elsewhere keeps 57 to 92% of the supported pixels at single-view rates of 0.6 to 19%, whereas VGGT's native confidence has no transferable scale.

The benchmark's own question. The benchmark defines unsupported geometry as a surface no input view shows; the vis protocol's positive class is a surface only one view shows. Put back to the benchmark's filled patches (run P6-vis-on-fill), the visibility heads score AUROC 0.70 to 0.75 on DTU and 0.62 to 0.67 on CO3Dv2, the controls 0.60 to 0.65, border and depth edges 0.43 to 0.51, all below native confidence on the same rows (run P6-vis-on-fill). A head trained to find single-view geometry does not fire on a patch every view shows, for the reason the reprojection and perturbation baselines do not: the invented surface is coherent across views and across the features. It is coherent across models too.

Disagreement between two backbones on the same views (depths aligned by a label-free median ratio, six pairs over VGGT, Pi3X, Pi3, DA3 and MapAnything; run P8-xmodel) scores AUROC 0.52 to 0.61 on the DTU patches and 0.49 to 0.63 on CO3Dv2 (supplement): five models of three families fill the hole with the same surface. The head certifies agreement between views, and a consistent invention has it. The detector answers "which pixels lack corroboration", not "which surfaces were invented". It is not, however, the same surface whichever pixels fill the hole. With the same hidden sphere filled three ways (gray, blur, noise) and the backbone run on each, the spread of the invented depth over the fills, aligned outside the hidden region, separates the patch from supported geometry at AUROC 0.82 to 0.98 on DTU and 0.62 to 0.96 on CO3Dv2 (run P9-refill; supplement), above native confidence for VGGT, Pi3X and DA3 and level with it for Pi3 on DTU, above it for VGGT, Pi3X and Pi3 and below it for DA3 on CO3Dv2. The median relative spread is 1.7 to 2.1% inside the hidden region against 0.1 to 0.6% outside it, about the invented depth's own error. Refilling the region in one view only moves the surface in that view and, an order of magnitude less, in the untouched views (supplement). The invention is coherent across views, features and models because all of them read the same filled pixels; change the pixels and it moves. A real capture offers no second fill, so this is a diagnosis rather than a detector, but it says where the invented surface comes from: the model reads the hole and draws what its pixels suggest, not a surface fixed by the prior.

8. Limitations and conclusion

The head certifies multi-view agreement, not truth: a surface every view agrees on passes, whether a camera saw it or the prior drew it, and the benchmark's filled patch is that case. The benchmark labels are a proxy for real occlusion, which the stress tests bound but do not remove. CO3Dv2 ground truth is COLMAP depth with 1 to 3% noise, and three backbones saw CO3Dv2 in training; the landmarks' ground truth is multi-view stereo with holes, so single-view labels there are a lower bound. MapAnything has only its native confidence on the vis protocol. Static scenes only; all findings depend on the released checkpoints.

When a feed-forward 3D model has not seen a surface, it mostly does not say so: a quarter to four fifths of the geometry it invents passes a confidence filter tuned to keep 90% of what it observed. The answer is not to stop the models from imagining but to make them say when they do. The models already know: a linear probe on their cross-view features separates single-view from supported geometry at AUROC 0.99 where native confidence scores 0.8, and a small head trained on the backbone's own multi-view agreement, with no ground truth, reads the same signal on four backbones and four datasets it never saw. What it cannot do sets the next problem: a surface the model invents consistently in every view passes every cross-view test we have, and five different models invent the same one; what moves it is the pixels it was drawn from, which a real capture does not let us change.

LLM Usage Statement

A general-purpose LLM assistant was used, under the authors' direction, for parts of the code (the benchmark runner, the model wrappers, the run scripts and figure scripts), for literature lookup, and for editing the text of this paper, including grammar. The study design, the definitions and metrics, the choice and freezing of every hyperparameter, the execution and checking of every experiment, the verification of every reported number against its logged run, and the conclusions are the authors' own, and the authors are responsible for all content. The assistant was not used to generate results, figures from unlogged data, or references; every citation was checked against the published record. The tool is named in the camera-ready version.

References

[1] David Ahmedt-Aristizabal, Mohammad Ali Armin, Russell Tsuchida, and Lars Petersson. GeoCond: A Conditioning-Aware reliability adapter for Feed-Forward 3D reconstruction. *arXiv:2609.18465*, 2026.

[2] Denis M. Akola and David F. Fouhey. Zero-Shot novel depth synthesis using 3D foundation models scene representations. *arXiv:2609.04174*, 2026.

[3] Jelena Bratulić, Sudhanshu Mittal, Thomas Brox, and Christian Rupprecht. On geometric understanding and learned priors in Feed-forward 3D reconstruction models. *arXiv:2512.11508*, 2025.

[4] Jinhyeok Choi, JoungBin Lee, Jongmin Lee, Minkyeong Jeon, Jiyoung Kim, Kaeun Han, Yoonjin Yoon, and Seungryong Kim. L3G-World: Real-time infinite world generation with latent 3D gaussian memory. *arXiv preprint arXiv:2610.40614*, 2026.

[5] Alibek Dadajonov and Hikmatillo Umarhojiyev. Sharper frames, worse rooms: Viewpoint coverage dominates image quality in feed-forward 3D reconstruction. *arXiv preprint arXiv:2610.15854*, 2026.

[6] Haoge Deng, Ting Pan, Fan Zhang, Yang Liu, Zhuoyan Luo, Yufeng Cui, Wenxuan Wang, Chunhua Shen, Shiguang Shan, Zhaoxiang Zhang, and Xinlong Wang. Uniform discrete diffusion with metric path for video generation. *arXiv:2510.24717*, 2025. URSA.

[7] Nanxing Nick Deng, Qing Cheng, Niclas Zeller, and Daniel Cremers. A calibration audit of confidence in Feed-Forward 3D reconstruction models. *arXiv:2608.29705*, 2026.

[8] Rasmus Jensen, Anders Dahl, George Vogiatzis, Engin Tola, and Henrik Aanæs. Large scale multi-view stereopsis evaluation. In *CVPR*, 2014.

[9] Lihan Jiang, Yucheng Mao, Linning Xu, Tao Lu, Kerui Ren, Yichen Jin, Xudong Xu, Mulin Yu, Jiangmiao Pang, Feng

Zhao, Dahua Lin, and Bo Dai. AnySplat: Feed-forward 3D gaussian splatting from unconstrained views. *arXiv preprint arXiv:2505.23716*, 2025.

[10] Yuhe Jin, Dmytro Mishkin, Anastasiia Mishchuk, Jiří Matas, Pascal Fua, Kwang Moo Yi, and Eduard Trulls. Image matching across wide baselines: From paper to practice. *International Journal of Computer Vision*, 129:517–547, 2021.

[11] Jinyoung Jun, Lei Chu, Jiahao Li, Yan Lu, and Chang-Su Kim. Perfecting depth: Uncertainty-Aware enhancement of metric depth. *arXiv:2506.04612*, 2025.

[12] Bingxin Ke, Anton Obukhov, Shengyu Huang, Nando Metzger, Rodrigo Caye Daudt, and Konrad Schindler. Repurposing Diffusion-Based image generators for Monocular Depth estimation. In *CVPR*, 2024.

[13] Nikhil Keetha, Norman Müller, Johannes Schönberger, Lorenzo Porzi, Yuchen Zhang, Tobias Fischer, Arno Knapitsch, Duncan Zauss, Ethan Weber, Nelson Antunes, Jonathon Luiten, Manuel Lopez-Antequera, Samuel Rota Bulò, Christian Richardt, Deva Ramanan, Sebastian Scherer, and Peter Kontschieder. MapAnything: Universal Feed-Forward metric 3D reconstruction. *arXiv:2509.13414*, 2025.

[14] Vincent Leroy, Yohann Cabon, and Jerome Revaud. Grounding image matching in 3D with MASt3R. In *ECCV*, 2024.

[15] Daisy Li, Kyle Gao, Quanyun Wu, Hanna Chomko, John S. Zelek, and Jonathan Li. When wider views fail: Stress-testing feed-forward 3D reconstruction. *arXiv preprint arXiv:2609.24839*, 2026.

[16] Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large Vision-Language models. In *EMNLP*, 2023.

[17] Haotong Lin, Sili Chen, Junhao Liew, Donny Y. Chen, Zhenyu Li, Guang Shi, Jiashi Feng, and Bingyi Kang. Depth Anything 3: Recovering the visual space from any views. *arXiv:2511.10647*, 2025.

[18] Sammam Mahdi, Fariha Binta Salim, Rakin Bin Rabbani, and Aniqua Nusrat Zereen. ReVar3R: Gauge-Aware perturbation uncertainty for Feed-Forward 3D reconstruction. *arXiv:2610.07883*, 2026.

[19] Fabian Mentzer, David Minnen, Eirikur Agustsson, and Michael Tschannen. Finite scalar quantization: VQ-VAE made simple. In *ICLR*, 2024.

[20] Tuan Duc Ngo, Chuang Gan, and Evangelos Kalogerakis. VolFill: Single-View amodal 3D scene reconstruction with volumetric flow matching. *arXiv:2605.31466*, 2026.

[21] Hoang Nguyen, Xiaohao Xu, and Xiaonan Huang. The 3D mirage: Probing and taming 3D hallucinations. *arXiv:2512.15423*, 2025.

[22] Jingyi Pan, Dan Xu, and Qiong Luo. Pose-free feed-forward 3D inpainting via learnable mask attention and support token refinement. *arXiv preprint arXiv:2610.11857*, 2026.

[23] Soumava Paul, Prakhar Kaushik, and Alan Yuille. Can these views be one scene? Evaluating multiview 3D consistency when 3D foundation models hallucinate. *arXiv:2605.18754*, 2026.

[24] Jeremy Reizenstein, Roman Shapovalov, Philipp Henzler, Luca Sbordone, Patrick Labatut, and David Novotny. Common objects in 3D: Large-Scale learning and evaluation of real-life 3D category reconstruction. In *ICCV*, 2021.

[25] Thomas Schöps, Johannes L. Schönberger, Silvano Galliani, Torsten Sattler, Konrad Schindler, Marc Pollefeys, and Andreas Geiger. A multi-view stereo benchmark with high-resolution images and multi-camera videos. In *CVPR*, 2017.

[26] Giuseppe Stracquadanio, Kevin Raj, Julia Grabinski, and Stefan Roth. ReconSplat: Generalizable 3D scene reconstruction beyond observed views. *arXiv preprint arXiv:2608.28895*, 2026.

[27] Alexandru Telea. An image inpainting technique based on the Fast Marching Method. *Journal of Graphics Tools*, 9(1), 2004.

[28] Adityan Thirumani Srinivasan and David Nordström. Does VGGT know what it does not know? *arXiv preprint arXiv:2610.35806*, 2026.

[29] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. VGGT: Visual geometry grounded transformer. In *CVPR*, 2025.

[30] Jianyuan Wang, Minghao Chen, Shangzhan Zhang, Nikita Karaev, Johannes Schönberger, Patrick Labatut, Piotr Bojanowski, David Novotny, Andrea Vedaldi, and Christian Rupprecht. VGGT-ω. *arXiv:2605.15195*, 2026.

[31] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. DUSt3R: Geometric 3D vision made easy. In *CVPR*, 2024.

[32] Yifan Wang, Jianjun Zhou, Haoyi Zhu, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, and Tong He. π^3: Scalable Permutation-Equivariant visual geometry learning. *arXiv:2507.13347*, 2025.

[33] Zihao Zhu, Wenyuan Zhao, Nuo Chen, Chao Tian, and Zhiwen Fan. Trust it or not: Evidential uncertainty for Feed-Forward 3D reconstruction with Trust3R. *arXiv:2605.19539*, 2026.

[34] Filippo Ziliotto, Luciano Serafini, Lamberto Ballan, and Tommaso Campari. What VGGT knows about overlap: Probing geometric foundation models for co-visibility. *arXiv preprint arXiv:2607.09503*, 2026.